Fan Zhang

Portfolio webpage: https://fanzhang0830.github.io
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Summary

Geospatial Data Scientist with a Ph.D. and over 5 years of experience in remote sensing, hydrology, raster/vector analytics, environmental modelling, GeoAI, and WebGIS-based decision-support workflows. Experienced in building Python-based geospatial pipelines for satellite imagery, high-resolution commercial imagery, in-situ observations, hydrological forecasts, feature engineering, model validation, spatial cross-validation, uncertainty analysis, and spatial prediction. Skilled in applying machine learning, image segmentation, object detection, CNN-based classification, deep learning tools, SAM, Hugging Face models, FastAPI, Docker, PostGIS, and GeoParquet to environmental, water-related, and cold-region monitoring problems.

Core Skills

- **GeoAI & Machine Learning**: Regression, classification, Random Forest, CNN-based classification, image segmentation, object detection, spatial cross-validation, uncertainty analysis, model validation, spatial prediction
- **Remote Sensing & Raster Analytics**: Sentinel-2, high-resolution commercial imagery, multispectral/red-edge features, NDVI/NDWI, band ratios, GeoTIFF, raster preprocessing, multi-temporal imagery
- **Geospatial Data Engineering**: GeoPandas, Rasterio, GDAL, xarray, PostGIS, GeoParquet, SQL, Google Earth Engine, geemap, raster/vector integration
- **Deep Learning Tools**: PyTorch, TensorFlow, Hugging Face, SAM, ArcGIS Pro deep learning tools
- **Deployment & Automation**: FastAPI, Docker, reusable Python modules, batch prediction workflows, API-based geospatial services, reproducible project structure
- **Domain Expertise**: Hydrological modelling workflows, flood forecasting, river ice analysis, water quality monitoring, environmental risk assessment, watershed analytics

Professional Experience

Remote Sensing Specialist, WSP E&I (Ottawa, Canada) Nov 2023 — Present

- Developed Python-based remote sensing and geospatial ML workflows for environmental projects using multi-source raster, vector, commercial imagery, and field datasets.
- Built classifications, regression, image segmentation, and feature extraction workflows to generate predictive spatial outputs and support environmental risk assessment.
- Conducted data cleaning, feature engineering, model validation, uncertainty review, documentation, and communication of spatial insights to project teams and stakeholders.

Climate Model Specialist, Geosapiens (Quebec City, Canada) Aug 2022 — Aug 2023

- Processed large-scale climate, hydrological, and geospatial datasets to support floodplain risk modelling and climate-risk assessment.
- Developed Python-based workflows for data preprocessing, exploratory analysis, feature generation, model calibration support, and result validation.
- Created geospatial visualizations and analytical outputs to interpret model results, identify key flood-risk variables, and support decision-making.

Education

University of Saskatchewan, Canada Sep 2015 – Aug 2020

Doctor of Philosophy

Research focus: remote sensing, hydrology/hydraulics, river ice analysis, and geospatial modelling

Projects

Chlorophyll-a Estimation from Sentinel-2 Imagery Using Machine Learning

Time: 2023 GitHub repo: github.com/FanZhang0830/chl_prj

- Built a Python-based remote sensing ML workflow using Sentinel-2 multispectral bands, red-edge bands, NDVI/NDWI, band ratios, and in-situ measurements for chlorophyll-a prediction.
- Performed raster preprocessing, feature extraction, satellite–field data co-location, model-ready sample generation, and multi-temporal image application across July–October 2023.
- Trained and validated a Random Forest regression model with R^2/RMSE/MAE, then applied the trained model to generate spatially continuous chlorophyll-a prediction rasters.

- Implemented reusable Python modules and geemap-based workflows for scalable Sentinel-2 image processing and raster-based environmental prediction.

River Ice Classification Using High-Resolution Commercial Imagery and Predictive Remote Sensing Features

Time: 2023-2025

- Built a cleaned river ice feature database from multi-source remote sensing datasets, associating observed ice classes with spectral bands, spatial attributes, and image-derived features.
- Applied ArcGIS Pro pre-trained deep learning models to high-resolution commercial imagery to extract and classify river ice features.
- Developed a feature prediction workflow to estimate future remote sensing band characteristics and support classification of upcoming imagery based on learned ice-feature relationships.
- Conducted raster/vector preprocessing, feature engineering, classification review, and GIS-based validation to support cold-region environmental monitoring and risk assessment.

EcoHydroAI – Geospatial Data Science Platform Prototype

Time: 2026 (in progress)   Demo link: https://fanzhang0830.github.io/map.html

- Built a modular geospatial data pipeline combining vector watersheds, raster imagery, weather time series, and hydrological forecasts for spatial intelligence workflows.
- Designed model-ready feature engineering components for watershed-level aggregation, environmental indicators, and remote-sensing-derived variables.
- Prototyped an API-first Python architecture to support scalable spatial analytics, future ML deployment, and automated geospatial intelligence generation.